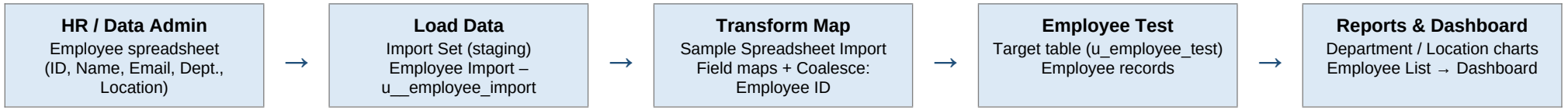
Project Design Phase-II

Data Flow Diagram & User Stories

Date	03 October 2026	**Team ID**	SWTID-2026-3942
Project Name	Import Data using Transform Maps	**Maximum Marks**	4 Marks

Data Flow Diagrams:

Employee data prepared in a spreadsheet is staged in an Import Set table, mapped to the Employee Test table by a Transform Map that uses Employee ID as its Coalesce field, and then summarised in reports on a dashboard.



Flow: Spreadsheet → Import Set (Employee Import) → Transform Map (field mapping + Coalesce) → Employee Test → Reports → Employee Analytics Dashboards.

Data Flow Description

- **Input:** Employee details are prepared in a spreadsheet and saved as an Excel (.xlsx) file.
- **Staging:** Load Data uploads the file into the Employee Import table; each spreadsheet row becomes one staged row.
- **Processing:** The Transform Map copies each staged column into its Employee Test field. With Coalesce on Employee ID, a row whose ID already exists updates that record and a new ID creates a new record.
- **Storage and validation:** Records are stored in Employee Test and checked through Transform History and the Import Set Row states.
- **Output:** Reports summarise employees by department and by location and list them; the three reports are placed on Employee Analytics Dashboards.

User Stories

The user stories below follow the project workflow and are the stories referred to in the UAT document (USN-1 to USN-11).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
HR / Data Admin	Data Collection	USN-1	As a data administrator, I can collect employee details (Employee ID, Name, Email, Department, Location) in one spreadsheet so that they can be imported together.	The spreadsheet has a header row and the five required columns.	High	Sprint-1
HR / Data Admin	Data Collection	USN-2	As a data administrator, I can upload the spreadsheet with Load Data so that its rows are held in an Import Set staging table.	The Employee Import table (u__employee_import) is created and all rows load without errors.	High	Sprint-1
HR / Data Admin	Data Preparation	USN-3	As a data administrator, I can clean the spreadsheet (consistent formats, no blank or repeated Employee IDs) so that the import runs without rejected rows.	Every row has a unique, non-empty Employee ID and consistent values.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
ServiceNow Admin	Data Preparation	USN-4	As a ServiceNow administrator, I can create the Employee Test table with the required fields so that imported records have a destination.	u_employee_test exists with Employee ID, Employee Name, Email, Department and Location.	High	Sprint-1
ServiceNow Admin	Transform Map	USN-5	As a ServiceNow administrator, I can create a Transform Map from Employee Import to Employee Test so that the staged rows have a defined route into the target table.	The map is saved with the correct source and target tables.	High	Sprint-1
ServiceNow Admin	Transform Map	USN-6	As a ServiceNow administrator, I can map the columns to target fields and set Employee ID as the Coalesce field so that repeated imports update employees instead of duplicating them.	Five field maps exist and Coalesce is true on Employee ID.	High	Sprint-1
HR / Data Admin	Transform Map	USN-7	As a data administrator, I can run the Transform Map so that spreadsheet rows are inserted or updated in the Employee Test table.	The transform completes with a Success code and the records appear in Employee Test.	High	Sprint-1
HR / Data Admin	Data Validation	USN-8	As a data administrator, I can open Transform History and the Import Set Rows so that I can see which rows were inserted, updated, ignored or rejected.	Row states and any error messages are visible for the import.	High	Sprint-2
HR / Data Admin	Data Validation	USN-9	As a data administrator, I can compare the spreadsheet row count with the Employee Test record count so that I know nothing was lost or duplicated.	The record count in Employee Test matches the number of unique employees in the spreadsheet.	High	Sprint-2
HR / Manager	Data Validation	USN-10	As an HR manager, I can view an employee list, employees by department and employees by location on one dashboard so that I can understand the workforce at a glance.	Employee Analytics Dashboards shows the three reports built on Employee Test.	Medium	Sprint-2
Team	Documentation	USN-11	As a team, we document the process with screenshots and test results so that the import can be repeated and reviewed.	Project documentation, UAT and performance-test documents are complete.	Medium	Sprint-2

Project Outcome

The project demonstrates loading spreadsheet data into an Import Set, mapping it to a target table with a Transform Map, preventing duplicates with Coalesce on Employee ID, validating the result through Transform History, and presenting it through reports and a dashboard.